



**RAJIV GANDHI COLLEGE OF ENGINEERING AND
TECHNOLOGY**

COMPUTER SCIENCE AND ENGINEERING

ASSIGNMENT - 2

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Reg No : 22TD0053

Subject Name : Object Oriented Analysis And Design

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Unit : 2

Teacher's Sign :

Department Vision And Mission

Vision

Cultivate student as technocrats researchers and entrepreneurs in the field of computer science and engineering.

Mission

- The impact quality education in computer science and engineering by means of learning techniques and value-added courses.
- To inculcate professional excellence and research culture by encouraging projects in cutting edge technologies, through industry interactions.
- To build leadership, skills ethical values and team work among the students.
- To strengthen the collaboration of departments and industry internships, mentor-ship and professional body activities.

PART - A

1) Differentiate state and Activity diagram.

State diagram

A statechart diagram shows the sequence of states that an object goes through during its life in response to outside stimuli and messages.

Activity diagram

An activity diagram is a variation or special case of a state machine, in which the states are activity representing the performance of operations and the transition are triggered by the completion of the operations.

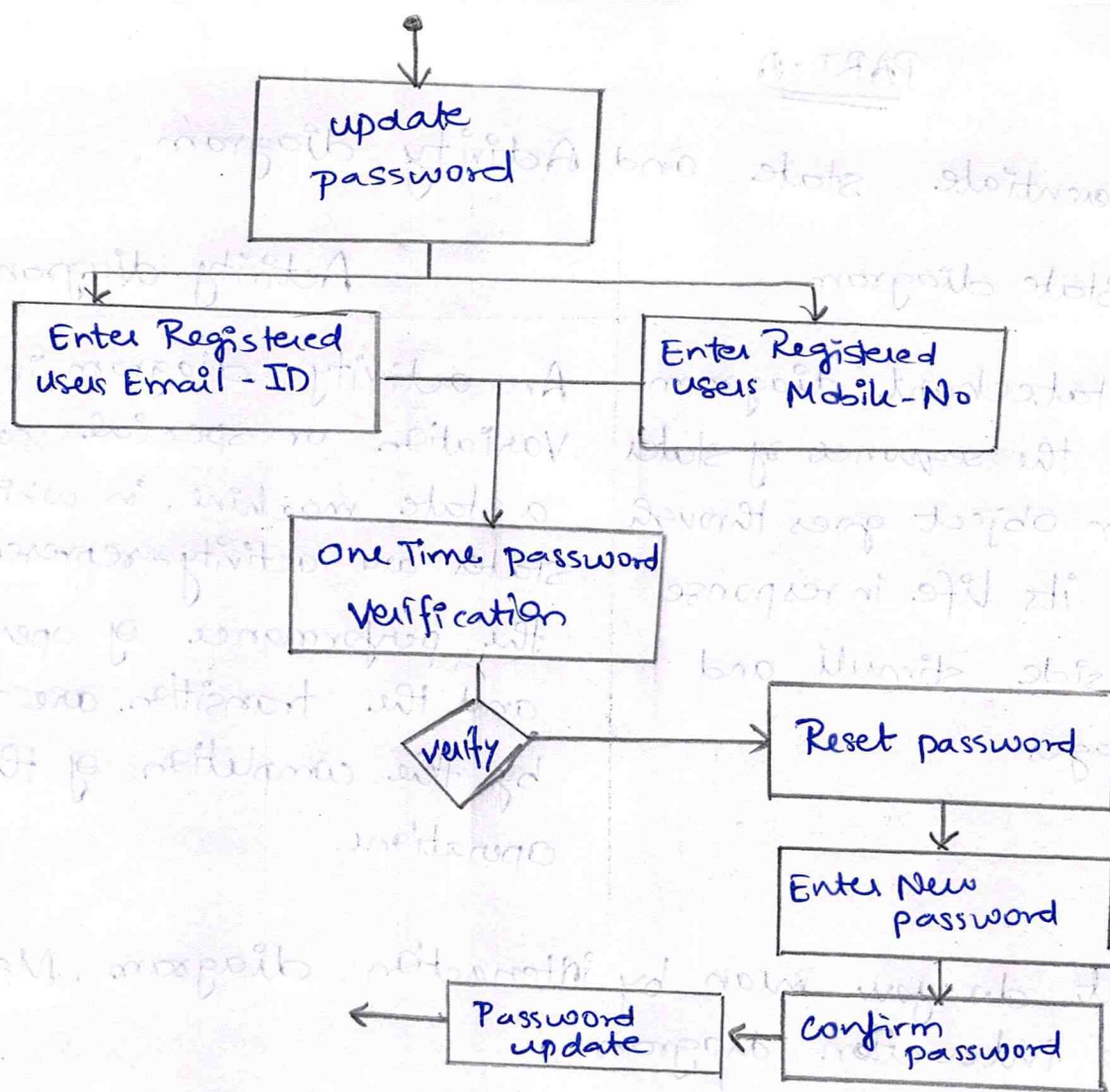
2) What do you mean by interaction diagram. Mention two interaction diagram.

Interaction diagrams are diagrams that describe how groups of objects collaborate to get the job done.

Interaction diagram captures the behaviours of a single use case, showing the pattern of interaction among objects.

Types: Sequence diagram
Collaboration diagram.

3) Draw an activity diagram that depicts the event while accessing the ATM to change passcode.



4) Differentiate use case and UML class diagrams.

Use case diagram

- Captures the functional requirements of a system
- Show the interaction between users and the system.
- Focuses on the system behaviour and functionality

UML class diagram

- Represents the static structure of a system.
- Chooses the classes and attributes and relationship between them.
- Focuses on the system's structure and organizations

5) List the three perspectives to apply UML.

1. Static perspective
2. Dynamic perspective
3. Functional perspective.

6) Distinguish between method and message in object

Method in object

- A procedure or function that is defined inside a class.

- Specifies the actions that an object can perform.

- Is invoked by the object itself or by another object.

Message in object

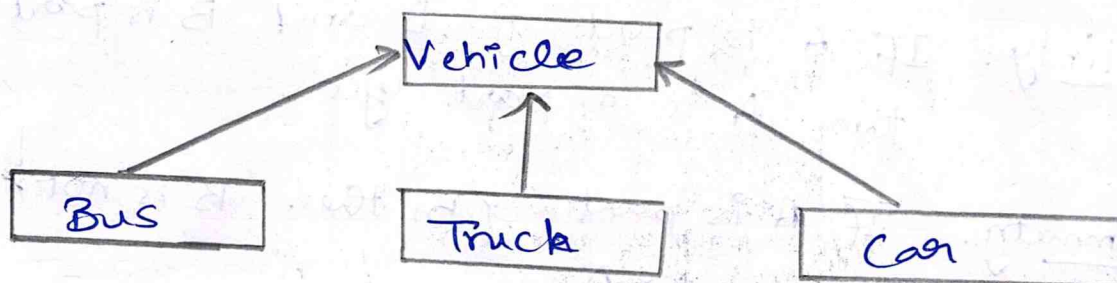
- A request or signal sent from one object to another.

- Specifies the action that the receiving object should perform.

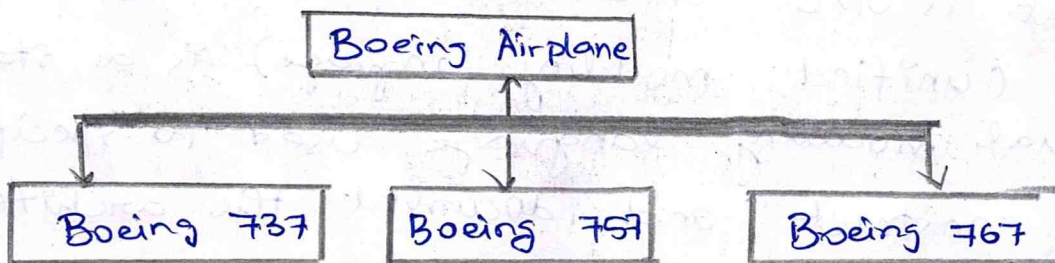
- Triggers the invocation of a method in the receiving object.

7) Draw a diagram to show logical connection between base and derived classes using relationship in UML.

Separate target style



Shared target style



Generalization is displayed as directed line with a closed, hollow arrowhead at the super-class end.

8) Compare Aggregation and composition.

Aggregation

- One object contains or manages a collection of other objects.
- The contained objects can exist independently of the container.

Eg: university (container)
and departments (contained objects)

Composition

One object is composed of other objects.

The contained objects cannot exist independently of the container.

Eg: Car (container) and
Engine (contained object)

9) List the relationship used in use cases.

- Transitivity
- Antisymmetry.

Transitivity: If A is part of B and B is part of C, then A is a part of C.

Antisymmetry: If A is part of B, then B is not part of A.

10) What is UML? Give the importance of UML.

UML (Unified Modeling Language) is a standardized visual modeling language used to specify, visualize, construct, and document the architecture of a software system.

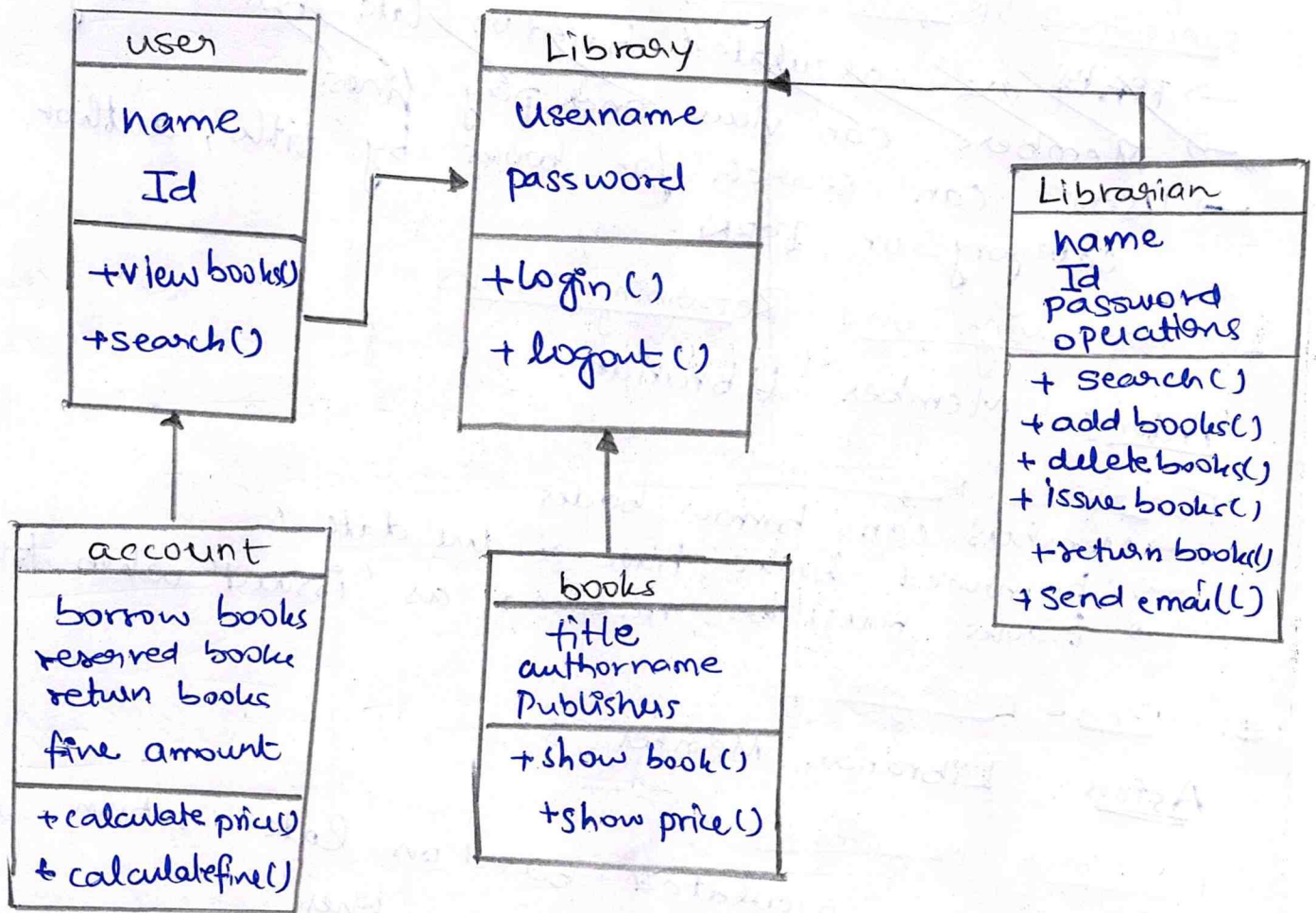
Importance of UML

- Standardization
- Visualization
- Reusability
- Scalability
- Reduced error.

Part - B

11) Develop a UML class, sequence and interaction diagram for library management system. State the functional requirements you are considering.

Class diagram



1. Uses Management

Actors: Librarian, Member,

Functional Requirements:

- Users must be able to register and login
- A librarian can manage members (add, update, delete)
- A member can update personal details.

2. Book Management

Actors: Librarian

Functional Requirements

- The system must allow adding, updating, and removing books.

→ Books must have unique identification (ISBN)

3. Book Search

Actors: Librarian, Member

Functional Requirements

- ~~Fines~~ are calculated based on late return day.
- Members can view and pay fines.
- Users can search for books by title, author, category or ISBN.

4. Borrowing and Returning books

Actors: Member Librarian.

Functional Requirements

- Members can borrow books
- Borrowed books have a due date.
- Books must be marked as 'issued' when borrowed

5. Fine management

Actors: Librarian, Member

Functional Requirement

- Fines are calculated based on late return day.
- Members can view and pay fines.

6. Reservation System

Actors: Member.

Functional Requirement

- Members can place reservation for unavailable books.
- Reserved books are allocated once available.

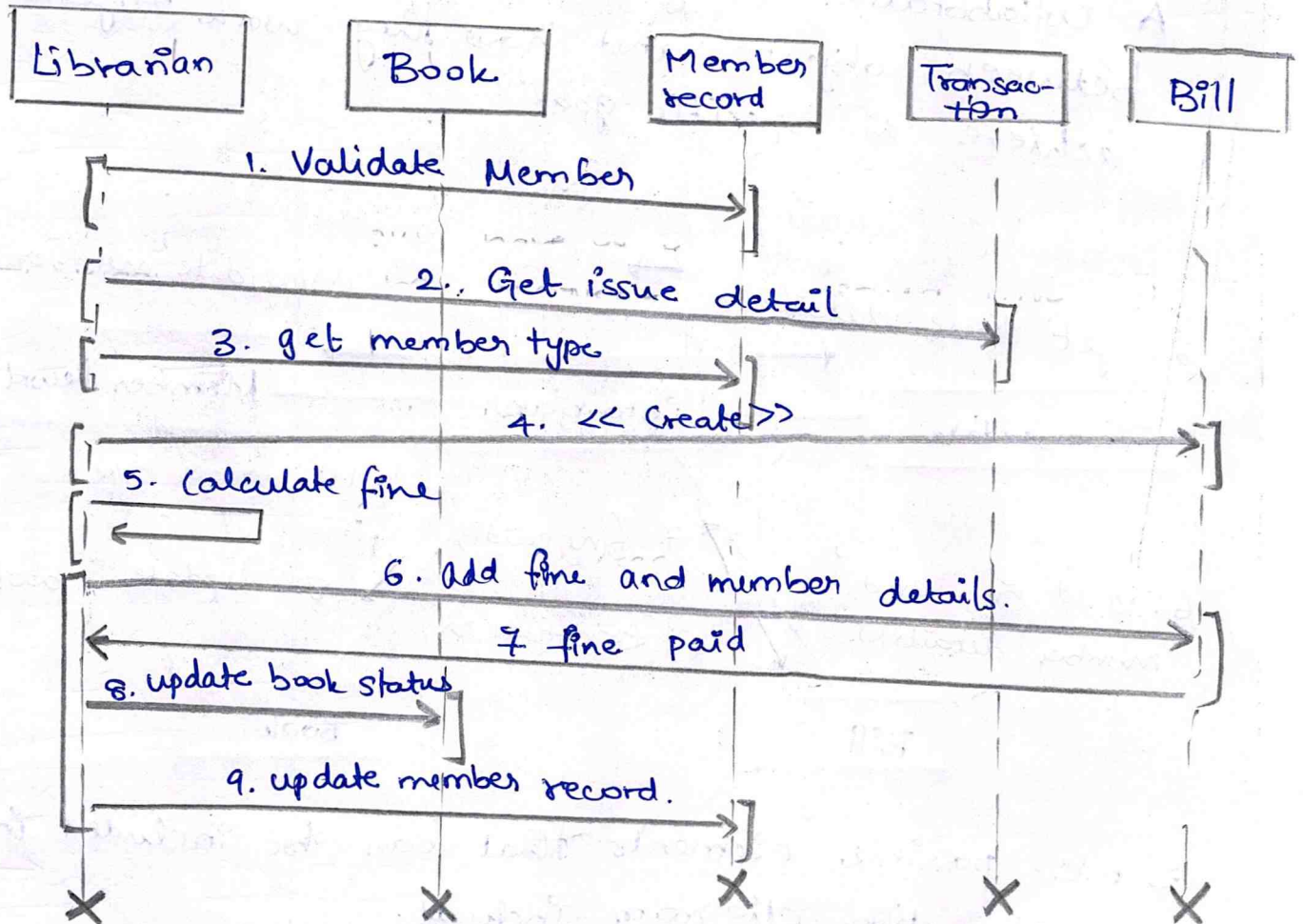
Sequence diagram

1. User login:

- User sends login request.
- ~~System~~ System verifies username and password.
- System sends login success / failure response.

2. Book borrowing

- User sends borrow book request.
- System checks book availability.
- System updates book status and user borrowing history.
- System sends borrow success / failure response.



3. Book return

- user sends return book request.
- System checks book status and user borrowing history.
- System updates book status and user borrowing history.

→ System sends return success / failure response.

4. Fine payment

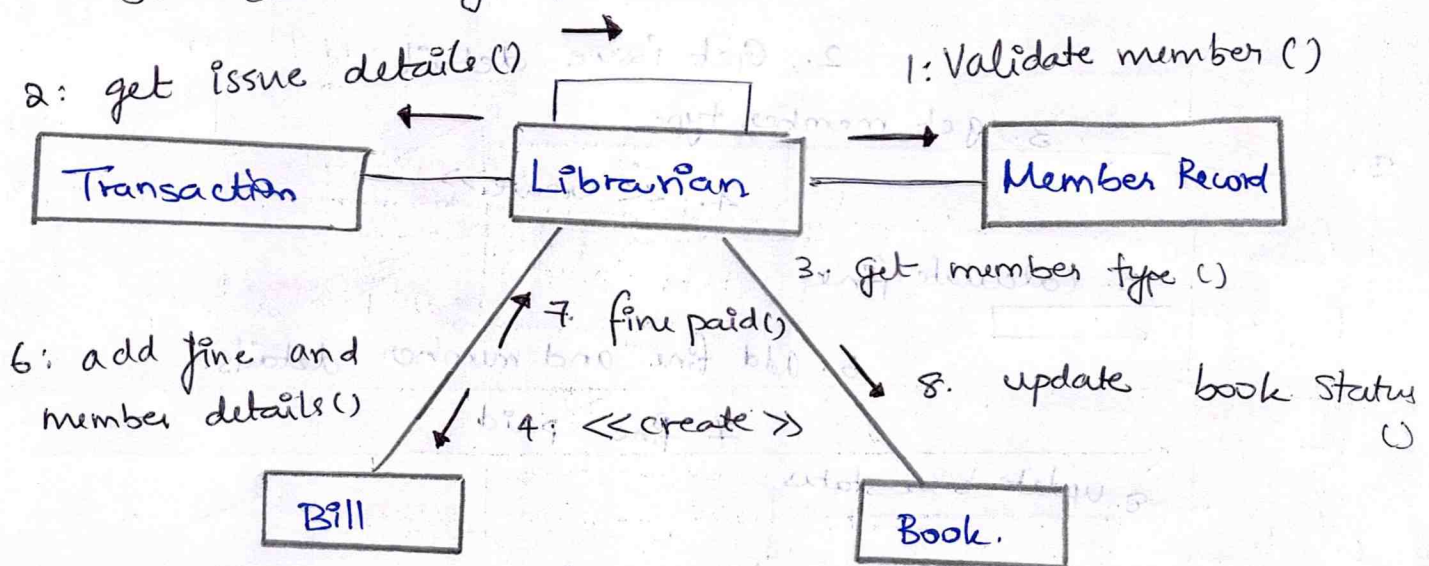
- User sends fine payment request
- System verifies payment details
- System updates fine status and user's account.
- System sends payment success / failure response.

A sequence diagram can illustrate the interaction between objects (eg: user, system, book) and the order in which these interactions occur.

Collaboration diagram

A collaboration diagram illustrates the interactions between objects and how they work together to achieve a specific goal.

5: Calculating fine ()



Some possible elements that can be included in a collaboration diagram include:

- ↳ Objects: (eg: user, Registration Manager, database)
- ↳ Messages: (eg: registration request, search request)
- ↳ Links (representing the relationships between objects)

↳ Roles (Representing the specific responsibilities of each object)

↳ Activation boxes (Representing the objects active state)

→ Lifelines (Representing the objects existence over time)

These classes can be further analyzed and referred to create UML class diagram for the library management system.

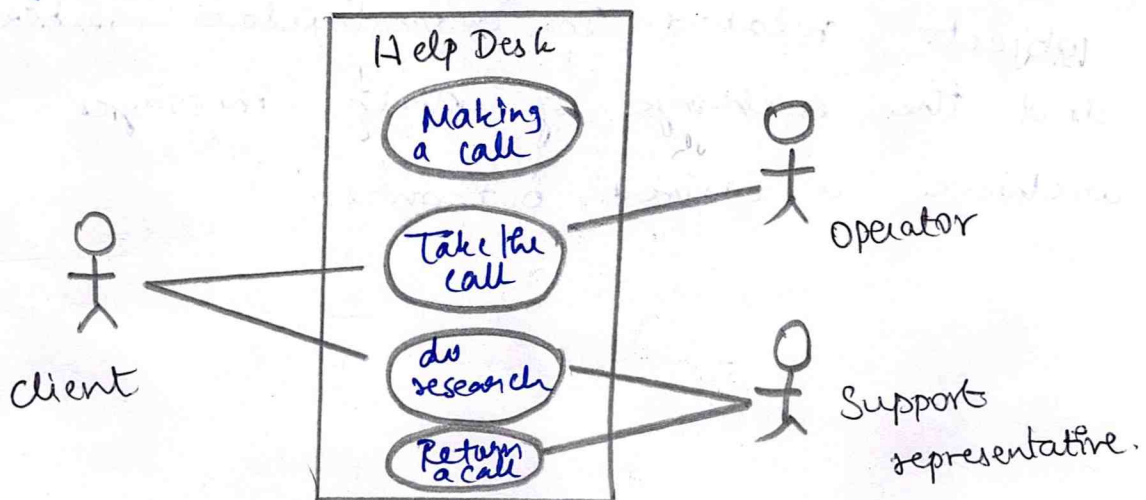
12) List any five structural UML diagram and brief them with appropriate illustrations.

The UML diagrams:

- Use case diagram
- UML Interaction diagram.
- UML Sequence diagram
- UML Collaboration diagram
- UML statechart diagram
- UML activity diagram.

Use case diagram

A use case diagram is a graph of actors, a set of use cases enclosed by a system boundary, communication association between the actors and the use cases, and generalization among the use cases.



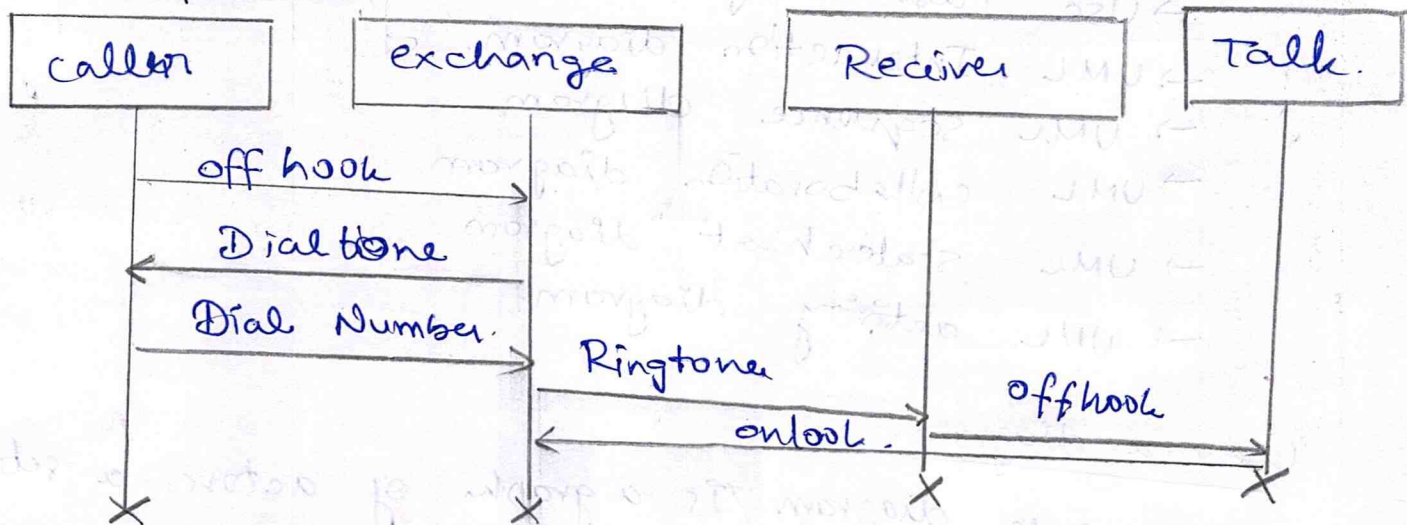
UML interaction diagram

- Interaction diagrams describe how groups of objects collaborate to get the job done.
- Interactions diagrams capture the behaviour of a single use case, showing the pattern of interaction among objects.

UML sequence diagram

- Sequence diagrams are an easy and intuitive way of describing the behaviour of a system.
- A sequence diagram shows an interaction arranged in a time sequence.

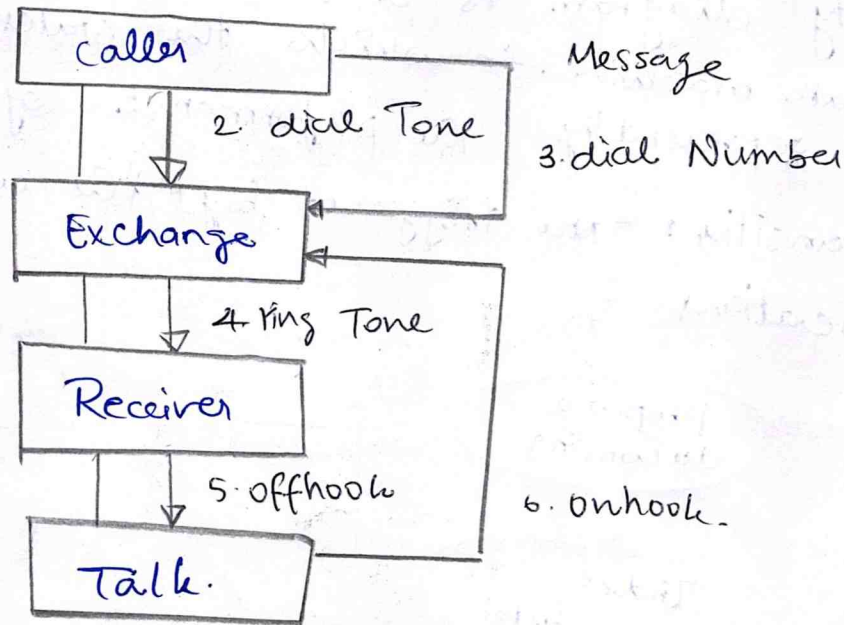
Telephone call



UML collaboration diagram

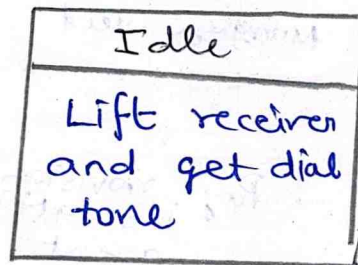
- A collaboration diagram represents a set of objects related in a particular context, and the exchange of their messages to achieve a desired outcome.

Telephone call



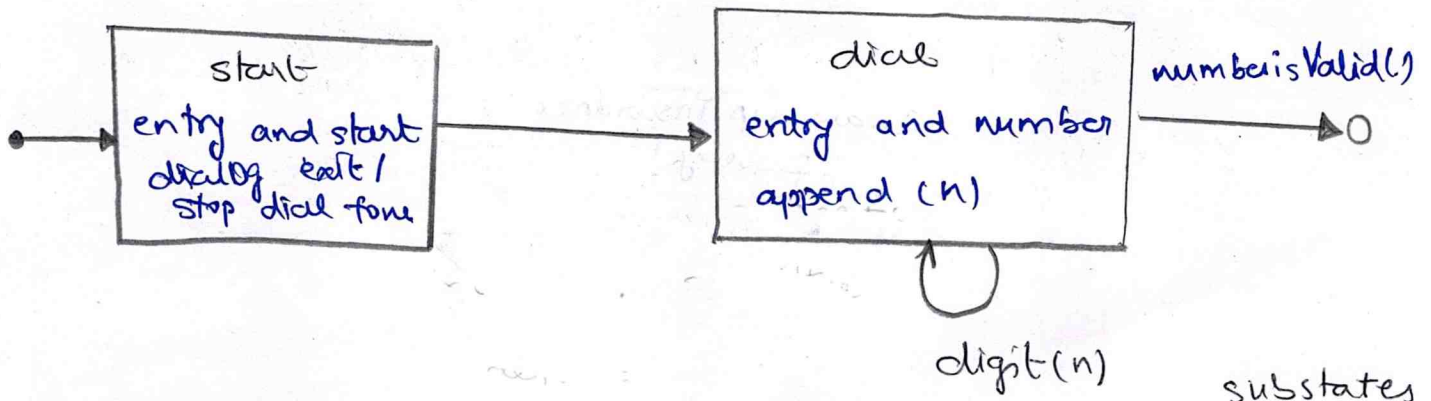
UML state chart diagram

A statechart diagram shows the sequence of states that an object goes through during its life in response to outside stimuli and messages.



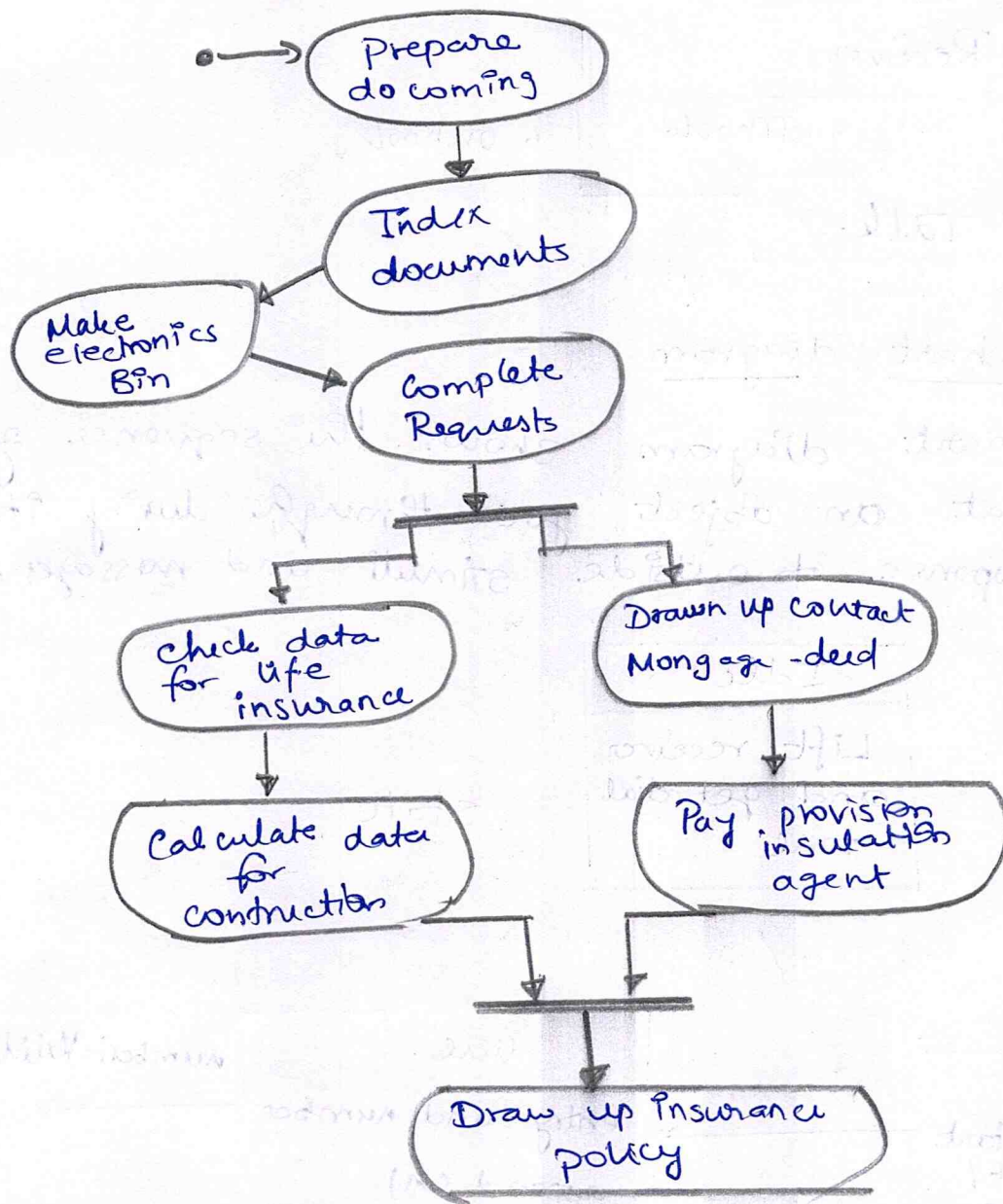
State

dialing



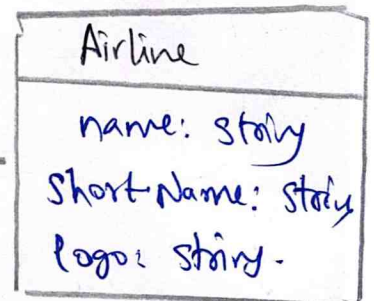
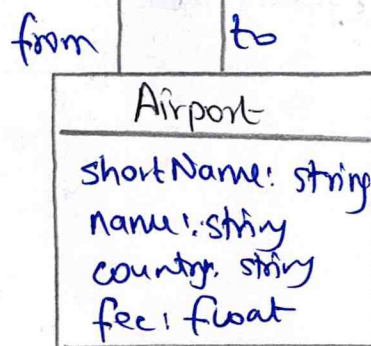
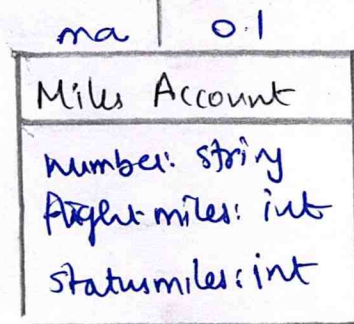
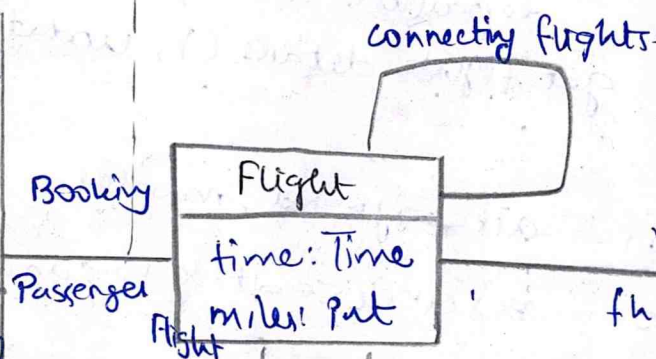
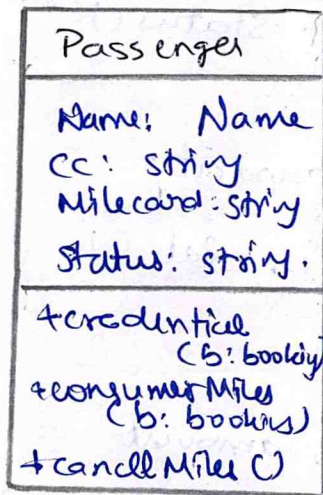
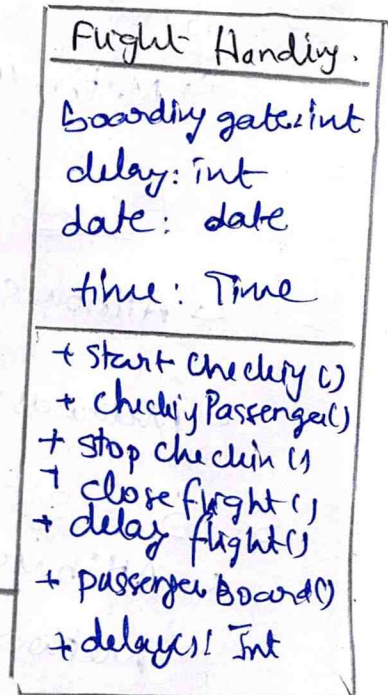
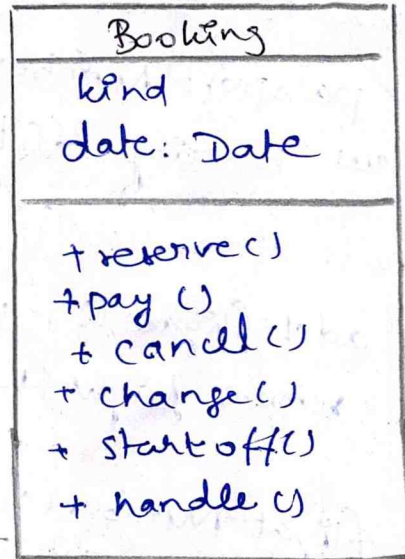
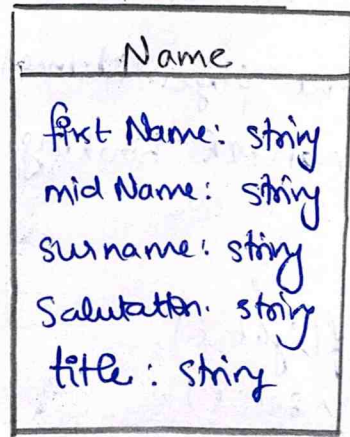
UML activity diagram

An activity diagram is a variation or special case of a state machine. in which the states are activities representing the performance of operations and the transitions are triggered by the completion of the operations.



13. Design the class diagram for airline Reservation System?
Find and draw conceptual classes for same.

Class diagram for airline Reservation System:



A class diagram for an airline Reservation system in object-oriented analysis and design typically includes major classes, their attributes, methods and relationships.

1. User:

↳ Attributes: userID, name, email, phone number

↳ Methods: register(), login(), update profile()

2. Passenger (inherits from user)

↳ Attributes: passport Number, frequent flyer Number.

↳ Methods: View Booking History(), cancel booking()

3. Admin (inherits from users)

↳ Methods: add flight(), update flight(),
remove flight(), manage users().

4. Flight

↳ Attributes: flight Number(), airline, source, destination
departure Time, arrival time, status

↳ Methods: get flight details(), update status()

5. Aircraft

↳ Attributes: aircraft Id, model, capacity

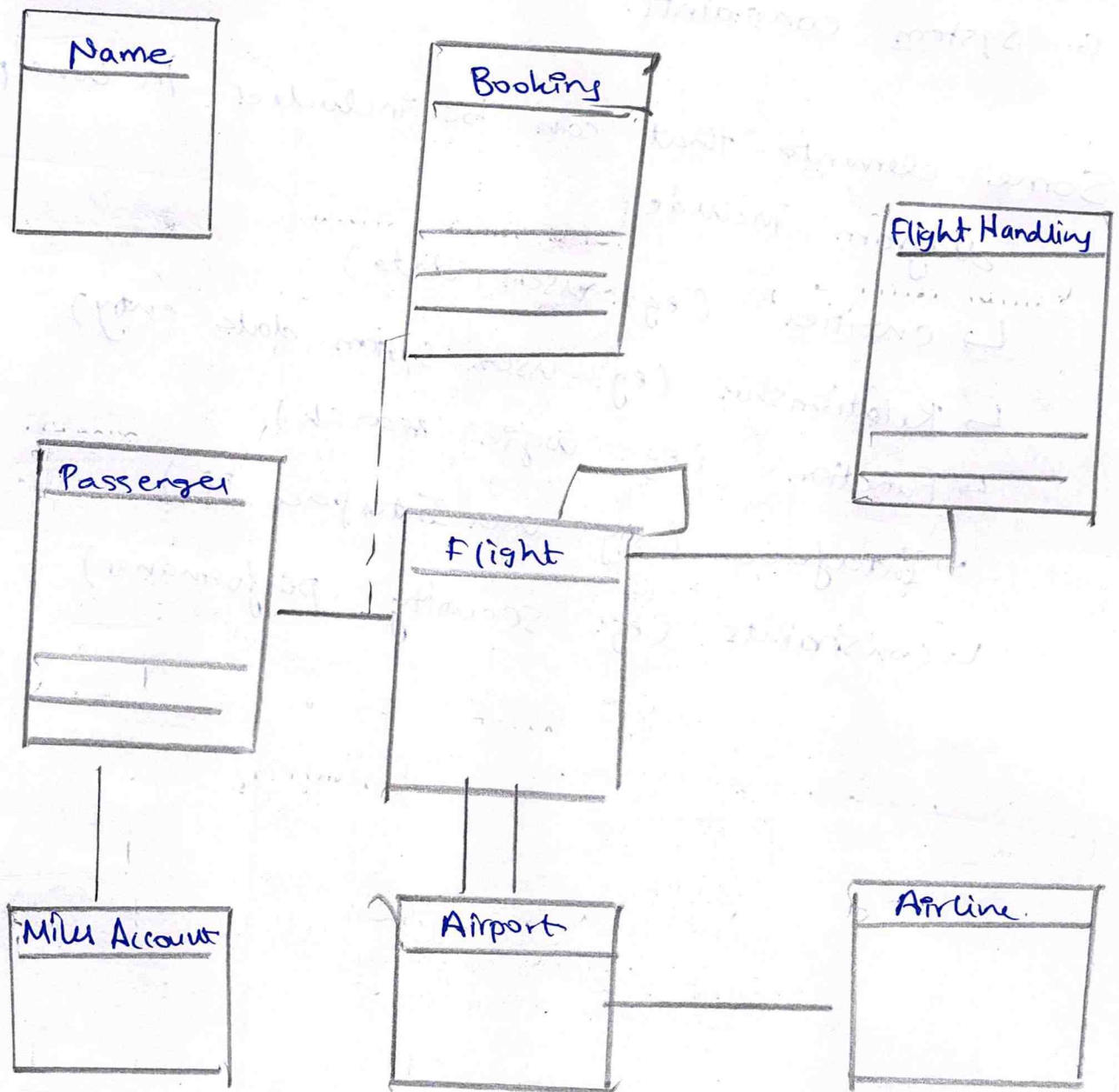
↳ Methods: reserve seat(), cancel seat().

6. Payment:

↳ Attribute: payment ID, booking ID, amount

↳ Methods: process payment(), refund payment()

Conceptual class diagram



A conceptual diagram, also known as a conceptual model, represents the overall concept and requirements of a system. Here are some functional requirements that can be represented.

1. System scope
2. User role
3. Functional Requirements.

4. Data Requirements.

5. System Interfaces

6. System constraints.

Some elements that can be included in conceptual diagram include.

↳ entities (eg: users, data)

↳ Relationship (eg: user, system, data-entry)

↳ Functions (eg: login search)

↳ Interface (eg: user interface, API)

↳ constraints (eg: security, performance)